

Product Review Sentiment Analyzer Using NLP

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Introduction

Product reviews play a significant role in influencing customer purchasing decisions on e-commerce platforms. With thousands of reviews being generated daily, manually analyzing customer opinions becomes difficult and time-consuming. The purpose of this project is to develop a Product Review Sentiment Analyzer that automatically classifies reviews into Positive, Negative, and Neutral sentiments using Natural Language Processing and Machine Learning techniques. The project is implemented using Python, Pandas, Scikit-Learn, TF-IDF Vectorization, LinearSVC, Streamlit, and Plotly.

Problem statement

The main objective of the Product Review Sentiment Analyzer is to help businesses and customers understand the sentiment behind product reviews. The application analyzes textual reviews, performs preprocessing, extracts features using TF-IDF, and predicts sentiment using a Linear Support Vector Classifier. A hybrid dataset consisting of Amazon product reviews and a synthetic review dataset was used for model training and evaluation. The system also provides a user-friendly Streamlit dashboard for real-time sentiment prediction.

Results and conclusion

Exploratory Data Analysis was performed to understand the distribution of sentiments and review patterns. The final dataset contained 11,597 reviews consisting of Positive, Negative, and Neutral sentiments. TF-IDF Vectorization was used for feature extraction and a LinearSVC model was trained for sentiment classification. **Accuracy: 87.97%. Precision: 88%. Recall: 87%. F1 Score: 88%.** The Streamlit dashboard successfully provides real-time sentiment prediction and sentiment distribution visualization.

Conclusion

A Product Review Sentiment Analyzer was successfully developed using Natural Language Processing and Machine Learning techniques. The LinearSVC model achieved an accuracy of 87.97%, demonstrating strong sentiment classification performance. The dashboard provides a simple and user-friendly interface for analyzing product reviews and predicting sentiments. The project highlights the practical application of NLP in customer feedback analysis and decision-making.

Reference

1. Scikit-Learn Documentation
2. Pandas Documentation
3. Streamlit Documentation
4. Plotly Documentation
5. Amazon Product Reviews Dataset